

A Practical Youla Gain-Tuning Framework for H_∞ Controller Realization

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Abstract: Standard H_∞ controller synthesis produces robust controllers with well-shaped sensitivity and complementary sensitivity transfer functions, S and T . However, at times H_∞ does not enforce strict requirements on sensitivity, in particular the desired requirement that T has unity gain at DC frequency. This results in typically negligible steady-state tracking error, as the H_∞ optimization produces $T(0) \approx 1$. In drive cycle applications where reference velocity profiles contain extended ramp segments, this negligible deviation is integrated over time into a growing, non-negligible bias. The conventional remedy is to augment the plant with an integrator prior to synthesis, but this increases the order of the plant model and can be inconvenient when the control designer's modeling has already been completed. This paper presents a post-synthesis gain adjustment method using Youla parameterization that corrects the DC tracking deficiency without modifying the plant or repeating H_∞ synthesis. The poles and zeros corresponding to the H_∞ controller's Youla transfer function Y are preserved, with a free parameter K replacing the gain of Y . Recalculating the controller after solving for the value of K that enforces $T(0) = 1$ results in a hybrid controller that retains the robustness of the original but with improved performance in ramp-input scenarios with minimal effort for the control designer. Simulation results on a vehicle speed tracking problem confirm elimination of accumulating bias while preserving robustness margins from the original H_∞ design.

Keywords: H_∞ control, Youla parameterization, gain tuning, control design automation

Received:

Revised:

Accepted:

Published:

Citation: Tan, R.; Yadav, S.; Assadian, F. A Practical Youla Gain-Tuning Framework for H_∞ Controller Realization. *Journal Not Specified* **2025**, *1*, 0. <https://doi.org/>

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1. Introduction

Robust control design incorporates concepts of model uncertainty, disturbances, and unmodeled dynamics in order to design controllers that ensure stability and performance despite uncertainty. Each source of potential degradation of control can be mitigated via loop-shaping, where the controller is designed to ensure that the model's closed-loop sensitivity, complementary sensitivity, and Youla transfer functions (S , T , Y respectively) have characteristics that ensure both robustness and performance, which are the following [3]:

- T is exactly 0 dB in low frequencies to track reference signals and low in high frequencies to attenuate high frequency unmodeled dynamics and sensor noise.
- S is low in low frequencies to attenuate disturbance and parameter variation effects.
- Y is low in high frequencies to limit actuator jitter resulting from sensor noise.
- The maximum magnitude of S is as close to 0 dB as possible, as this relates to system robustness.
- The maximum magnitude of Y is as low as feasible, as this determines the required actuator effort.

In H_∞ synthesis, the above characteristics are achieved by finding the controller that minimizes the maximum magnitude of any entry in the vector $[W_p S, W_d T, W_u Y]^T$ across all frequencies. The W_p , W_d , and W_u weighting transfer functions are approximately the inverse of the desired shapes for S , T , and Y respectively with regards to the goals of loop-shaping, thereby giving the H_∞ synthesis algorithm a cost function that naturally targets the desired loop shapes.

This makes the technique of H_∞ synthesis a powerful framework for constructing controllers that minimize worst-case gain from disturbances to regulated outputs [1,2]. For the control designer, H_∞ synthesis is a convenient method of indirectly designing robust controllers. Once a plant model G_P has been determined, the control designer tunes the controller by adjusting the cost weighting functions W_p , W_d , and W_u , representing the designer's goals for the closed-loop performance, both in achieving the desired overall loop shapes for robustness and performance as well as the specific crossover frequency, transfer function slopes, and magnitudes at specific frequency ranges.

While it can be convenient to adjust weights rather than the controller directly, the output of H_∞ synthesis does not strictly enforce loop-shaping goals since it only is aware of its cost function. The goal under current consideration is that the complementary sensitivity function, which, among other metrics, describes the ratio of the regulated output to the input reference signal, has unity gain at zero frequency (DC gain), or $T(0) = 1$ [3]. For many cases, the output of H_∞ synthesis results in $T(0) \approx 1$, for example $T(0) = 0.99985$ as observed for the system in [4]. Depending on the scenario, $T(0)$ can be either slightly below 1.0, resulting in undershoot, or slightly above 1.0, resulting in overshoot. Using the example of $T(0) = 0.99985$, this results in a steady-state error of only 0.015% for a step input which is often more than acceptable. However, in cases such as the drive cycle input used in [4], a reference signal that includes prolonged periods of ramping input results in the 0.015% undershoot being integrated and resulting in the regulated output drifting over time.

Fine-tuning the H_∞ output can be inconvenient, requiring an iterative process to tweak the input weights or requiring modifications to the plant model. The typical method used to enforce $T(0) = 1$ would be to augment the plant model with an integrator [5], naturally enforcing $T(0) = 1$. However, the selection of weighting functions for the H_∞ solver must be non-trivially adjusted to account for the effect of the integrator on the plant model, where the compensation on the weighting functions changes depending on the dynamics of the original plant. This introduces a second layer of iteration if the augmentation is performed later in the controller design workflow. In addition, augmentation increases the order of the plant which will typically increase the order of the resulting controller as well, if the order of weighting functions is held constant.

A relevant iterative Youla parameterization design method was implemented in [6], where the flexibility of Youla parameterization is utilized in order to perform loop-shaping at specific frequencies of the system. The iterative method is used in order to tune high-order controllers by modifying the existing, or previous-iteration, controller in a manner that maintains stability and achieves the desired closed loop transfer function characteristics.

The method described in this paper is a post-synthesis adjustment to the H_∞ output using Youla parameterization techniques that allows for strict enforcement of $T(0) = 1$ in an intuitive, easy to implement manner that is well-adapted to implementation in programming languages. Part of the benefit of our method is that the post-synthesis adjustment only requires a single scalar Youla shaping parameter and can be performed in a single step, compared to the iterative process of [6], making the adjustment of H_∞ controllers simple to implement. The hybrid method is referred to here as **Youla-H synthesis**.

When viewed from the perspective of controller design with Youla parameterization from scratch, it is much easier with Youla-H to achieve a robust controller when starting with the output of H_∞ rather than attempting to select a pole/zero structure with no prior information.

2. Methods

2.1. Notation

The standard transfer functions [3] used throughout are defined below.

- G_P —plant transfer function.
- G_C —controller transfer function.
- $L = G_C G_P$ —open-loop transfer function/return ratio, represents open-loop system response to reference input.
- $T = L/(1 + L)$ —complementary sensitivity transfer function, relates to closed-loop reference tracking relative to reference input.
- $S = 1/(1 + L)$ —sensitivity transfer function, relates to closed-loop disturbance and plant parameter variation sensitivity.
- $Y = G_C/(1 + L)$ —Youla transfer function, relates to closed-loop actuator effort relative to reference input.
- W_p —cost weight transfer function on S for H_∞ .
- W_d —cost weight transfer function on T for H_∞ .
- W_u —cost weight transfer function on Y for H_∞ .

The subscripts used throughout are:

- H—corresponds to H_∞ transfer functions.
- YH—corresponds to hybrid Youla-H transfer functions.

2.2. Gain Tuning

Equation 1 describes the generalized Y transfer function that would be output by the initial H_∞ synthesis in zero-pole-gain (ZPK) form.

$$Y_H(s) = K_H \frac{\prod_{i=1}^n (s + z_i)}{\prod_{j=1}^m (s + p_j)} \quad (1)$$

The hybrid Youla-H Y transfer function is given in Equation 2, where K_{YH} is a free scalar parameter that has not yet been determined, but the values of z_i and p_i are the same as in the H_∞ case.

$$Y_{YH}(s) = K_{YH} \frac{\prod_{i=1}^n (s + z_i)}{\prod_{j=1}^m (s + p_j)} \quad (2)$$

Alternatively, Y_{YH} can be described in terms of Y_H , K_H , and K_{YH} .

$$Y_{YH}(s) = \frac{K_{YH}}{K_H} Y_H(s) \quad (3)$$

Note that since $T = L/(1 + L)$ and $Y = G_C/(1 + L)$, $T = G_P Y$. Equation 4 shows the enforcement of strict tracking performance.

$$T_{YH}(0) = 1 = G_P(0) Y_{YH}(0) = \frac{K_{YH}}{K_H} Y_H(0) G_P(0) = K_{YH} G_P(0) \frac{\prod_{i=1}^n z_i}{\prod_{j=1}^m p_j} \quad (4)$$

This resolves into equation 5, setting the exact numeric value for K_{YH} .

$$K_{YH} = \frac{K_H}{Y_H(0)G_P(0)} = \frac{1}{G_P(0)} \frac{\prod_{j=1}^m p_j}{\prod_{i=1}^n z_i} \quad (5)$$

Setting K_{YH} in this manner algebraically enforces $T(0) = 1$ and, by extension, $S(0) = 0$. Since $G_C = Y/S$, this indicates that $G_C(0) \rightarrow \infty$ which corresponds to at least one pure integrator in the final controller.

This method assumes $G_P(0) \neq 0$ and $Y(0) \neq 0$. If $G_P(0) = 0$, the plant has zero DC gain, making $T(0) = G_P(0)Y(0) = 0$ for any finite K_{YH} . Strict DC tracking is then unachievable through gain adjustment alone, and the Youla-H method does not apply. If $Y(0) = 0$, the H_∞ -synthesized Youla transfer function contains a pure differentiator, which again causes $T(0) = G_P(0)Y(0) = 0$ for any finite K_{YH} . In either case, the designer must revisit the plant model, weighting function selection, or the entire control methodology before potentially applying the Youla-H method.

3. Plant Models

The first-order plant shown in Figure 1 and Equation 6 is that used in [4], which was used to design drive controllers for a non-linear vehicle drivetrain model.

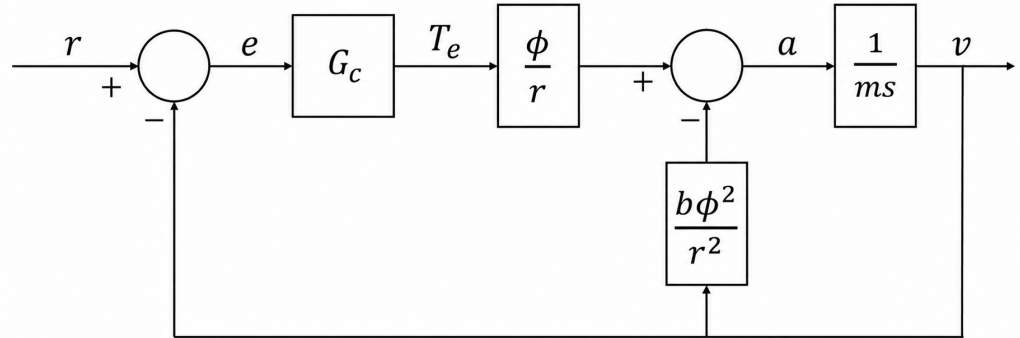


Figure 1. First-order linear plant model, controller, and feedback loop.

Figure 1 shows the controller G_C , which is not part of the plant but that outputs the engine torque T_e , being transformed through the gear ratio, ϕ , and the tire radius, r , into the driving acceleration on the vehicle. This acceleration is dampened by the $b\phi^2/r^2$ term, which represents the effect of motor friction transformed through the drivetrain. Finally, the net acceleration, a , is divided by the effective vehicle mass, m , and integrated in order to output the vehicle velocity, v , which is the tracked signal in the system. This linear plant model is a simplified subset of the dynamics that are modeled in the full non-linear simulation. This simplified model is used only to design controllers which, due to the robust nature of loop-shaping methods, still produces effective controllers for the nonlinear simulation used in [4]. The torque conversion, motor friction conversion, and mass / integration blocks of Figure 1 comprise the plant, G_P , shown in Equation 6.

$$G_P = \frac{v}{T_e} = \frac{\frac{\phi}{mr}}{s + \frac{b\phi^2}{mr^2}} \quad (6)$$

For the purposes of this paper, the vehicle parameters chosen are the scaled-down parameters for the test model built in [4]. The plant numerically reduces down to Equation 7.

$$G_P = \frac{22.624}{s + 0.0000432} \quad (7)$$

Note that this is an exceedingly simple plant model. Despite this, the H_∞ synthesis was not observed to produce exactly $T(0) = 1$ for any of the combinations of weighting functions that were tested, though it was not a rigorous search. As mentioned in the Introduction, a common method of compensating for the error in reference tracking is to augment the plant state space such that a pure integrator is produced by the controller. Instead, the hybrid Youla-H method is used in this paper in order to enforce strict tracking as a simple-to-implement alternative. If the Youla-H method had been utilized at the beginning of controller design, the weight selection process would have been much faster since exploration to find a controller with a better $T(0)$ value would not have been necessary. Further, work considering augmenting the plant with an integrator would not be necessary either.

At the end of the Results section, a second plant will be used to demonstrate a scenario where the Youla-H method applies even with an easier H_∞ optimization. See Section 4.5 for details.

4. Results

Bode plots are used to inspect the closed-loop transfer functions of the original H_∞ -produced controller and the hybrid Youla-H controller, using the linear plant transfer function. Afterwards, tracking error results are shown produced from the same nonlinear drivetrain model used in [4].

4.1. Closed-Loop Inspection

Figure 2 shows the T , S , and Y transfer functions for the H_∞ controller along with the inverses of the weighting functions W_d , W_p , and W_u . The crossover frequency for the W_d and W_p weighting functions is 18rad/s , corresponding to the control design performed in [4]. For this system, this weighting crossover frequency results in the final T and S crossover frequency being approximately 24rad/s , the consistent bandwidth chosen for all controllers in [4] and which is several times faster than the main input frequencies from the drive cycle reference input.

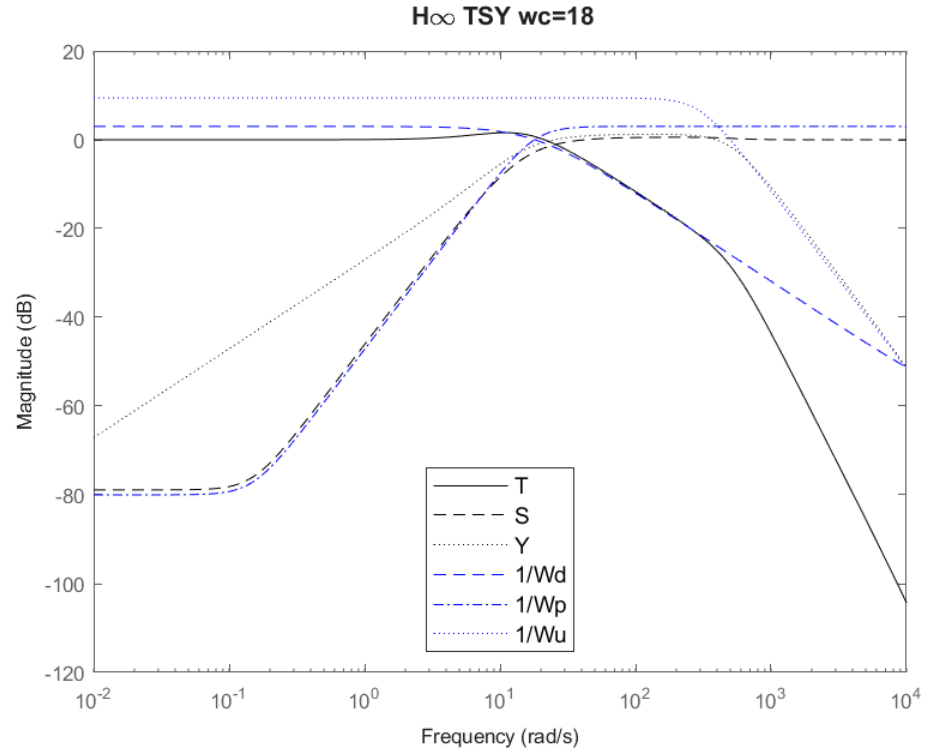


Figure 2. Closed-loop transfer functions for the H_∞ -synthesized controller.

For the drive train model and the weights selected, the H_∞ synthesis from [4] achieved $\gamma = 1.15$ which is higher than the ideal value of $\gamma < 1.00$. This is primarily due to the second-order weighting used on the low-frequency end of W_p and on the high-frequency side of W_u , which increases the optimization difficulty but is necessary to produce the desired degrees of sensor noise and disturbance attenuation to outperform other controllers. This sub-optimal γ value weakens the guarantees of H_∞ synthesis but is sufficiently low to still provide appropriate loop shaping, as evidenced in Figure 2. For the T shown in Figure 2, $T_H(0) = 0.99985$, a deficiency of 0.015%.

Figure 3 shows the T , S , and Y transfer functions for the gain-adjusted Youla-H controller. By inspection, it is very similar to the H_∞ Bode plot, with the primary visible change being that S continues to trend downwards with low frequency compared to the S in Figure 2, which instead plateaus at low frequency.

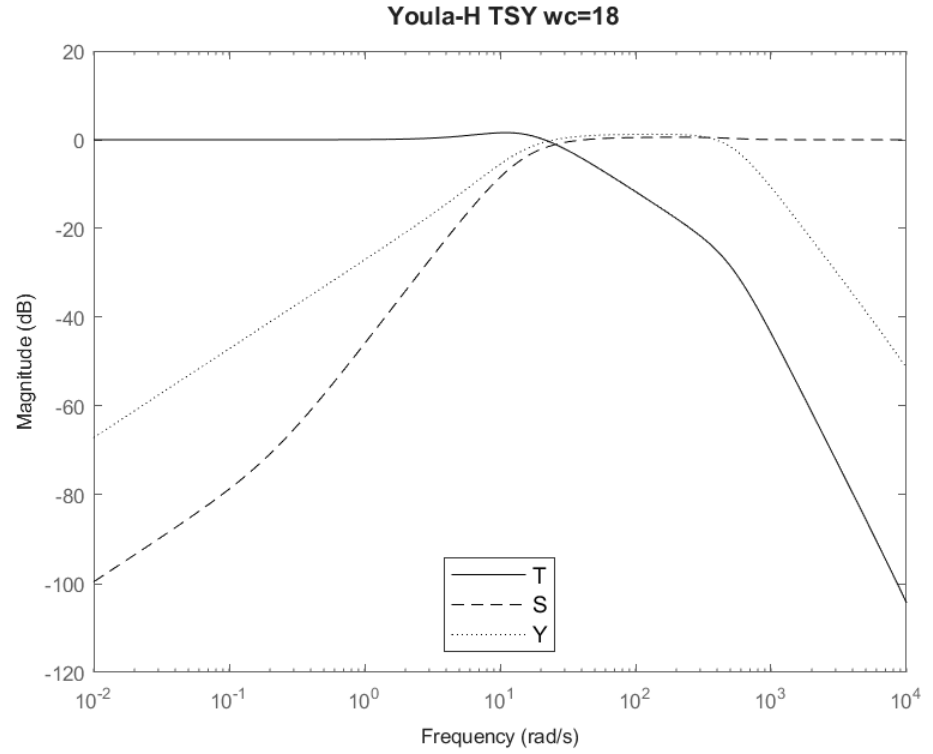


Figure 3. Closed-loop transfer functions for the Youla-H controller.

Since $S + T = 1$, the reduction in S at low frequency corresponds to an increase in T . We then observe that $T_{YH}(0) = 1.00000$.

Most strikingly, Figure 4 compares controller transfer functions, demonstrating that the very small numerical adjustment of $T(0)$ results in significantly different controller characteristics at low frequencies. This is because $G_C = Y/S$, and as $T(0) \rightarrow 1.0$, $S(0) \rightarrow 0$, resulting in the infinite DC gain at $G_C(0)$ corresponding to a pure integrator.

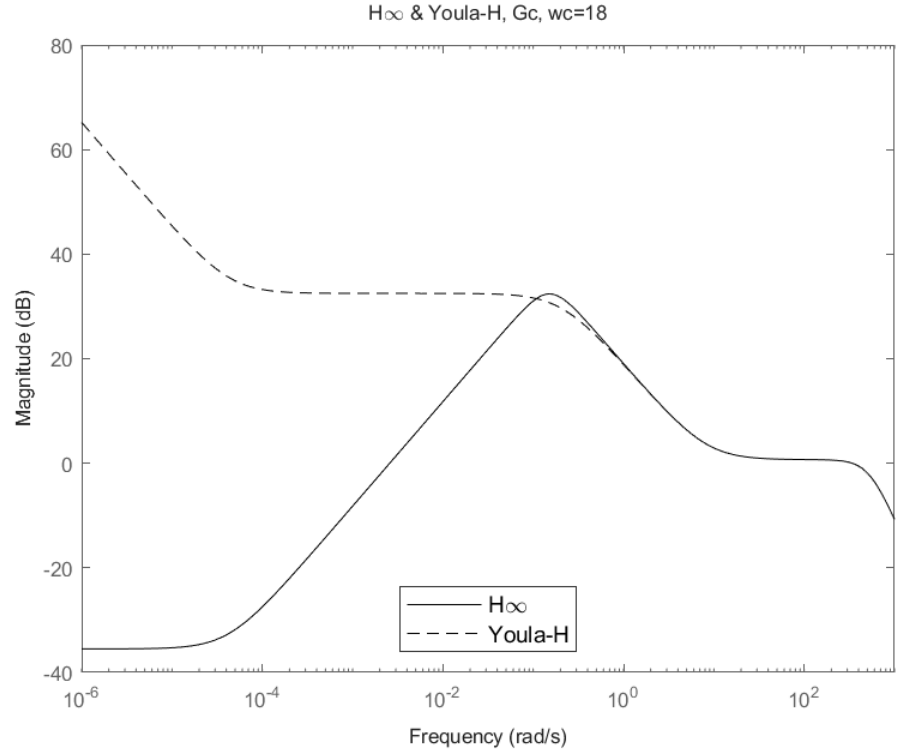


Figure 4. Bode magnitude plot of controller transfer functions G_C for both H_∞ and Youla-H.

Part of the reason for the deficiency in $T(0)$ is that the optimization goal of $\gamma < 1.0$ was not met. Due to the control design requirements of our specific system, $\gamma = 1.15$ was the best observed outcome. A benefit of the hybrid Youla-H gain tuning method under review is that sub-optimal γ values can still result in controllers that satisfy all requirements without having to change weighting functions. However, section 4.5 showcases that the Youla-H method can still be relevant even with $\gamma < 1.0$.

4.2. Transfer Functions

Equation 8 shows the Youla transfer function for the H_∞ -synthesized system.

$$Y_H = 4.5781 \frac{(s + 8.225)(s + 4.316 \times 10^{-5})(s^2 + 5.045 \times 10^4 s + 1.273 \times 10^9)}{(s + 1.926 \times 10^4)(s^2 + 25.89s + 216)(s^2 + 721.8s + 2.607 \times 10^5)} \quad (8)$$

Per the Youla-H method described in this paper, the poles and zeros of this Y are retained when determining the new hybrid Y . The 4.5781 DC gain is replaced with a variable DC gain K that is solved in order to ensure $T(0) = 1$. The result is the Youla transfer function for the Youla-H system shown below in equation 9.

$$Y_{YH} = 4.5787 \frac{(s + 8.225)(s + 4.316 \times 10^{-5})(s^2 + 5.045 \times 10^4 s + 1.273 \times 10^9)}{(s + 1.926 \times 10^4)(s^2 + 25.89s + 216)(s^2 + 721.8s + 2.607 \times 10^5)} \quad (9)$$

In performing the Youla-H method, the DC gain for Y is shifted by +0.0114%, from 4.5781 to 4.5787. This small change results in the complete removal of accumulating tracking bias with inputs that include ramping profiles.

Equations 10 and 11 show the resulting controller transfer functions for the H_∞ and Youla-H controllers, respectively.

$$G_{CH} = 4.5781 \frac{(s + 8.225)(s + 4.316 \times 10^{-5})(s^2 + 5.045 \times 10^4 s + 1.273 \times 10^9)}{(s + 1.926 \times 10^4)(s^2 + 0.2141s + 0.02291)(s^2 + 747.5s + 2.795 \times 10^5)} \quad (10)$$

$$G_{CYH} = 4.5787 \frac{(s + 8.225)(s + 4.316 \times 10^{-5})(s^2 + 5.045 \times 10^4 s + 1.273 \times 10^9)}{s(s + 1.926 \times 10^4)(s + 0.2113)(s^2 + 747.5s + 2.795 \times 10^5)} \quad (11)$$

The DC gain for both G_C transfer functions are the same as for their Y counterparts. The difference in dynamics between the two controllers is that the low-frequency complex pole pair in G_{CH} is split into a pure integrator and another low-frequency real pole in G_{CYH} .

4.3. Error Comparison

The error results correspond to tracking performance on the FTP75 drive cycle with the nonlinear drivetrain model. The FTP75 reference input is shown in Figure 5.

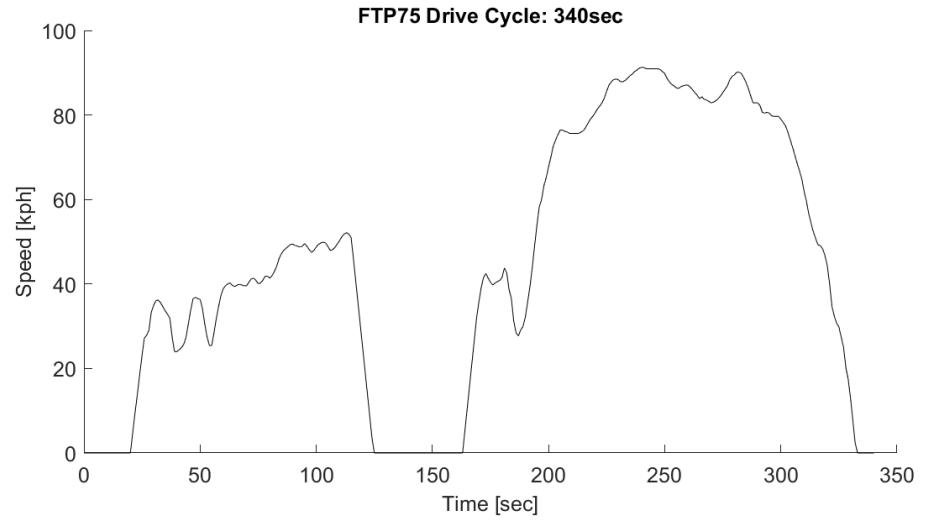


Figure 5. FTP75 drive cycle.

Figure 6 shows the normalized velocity tracking error of the H_∞ and Youla-H drive controllers on the nonlinear plant while tracking the FTP-75 drive cycle.

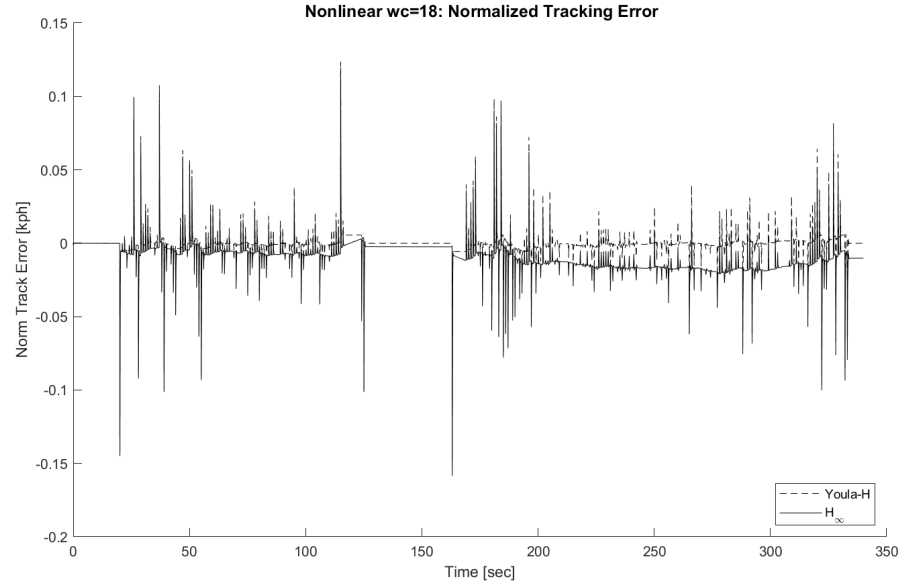


Figure 6. Tracking error of H_∞ and Youla-H drive controllers on nonlinear model.

Due to the FTP-75 drive cycle consisting of ramping inputs, the marginal $T_H(0)$ deficiency appears in the tracking results as an accumulating undershoot, while the Youla-H controller has no bias.

Note that the controller bandwidth is high relative to the inputs and disturbances seen in the nonlinear simulation, resulting in spikes of actuator effort and tight tracking. In [4], this bandwidth was selected across controllers in order to ensure tracking performance and due to the scaled simulation representing a high-power electric vehicle capable of high acceleration. In this paper, the high bandwidth helps to illustrate the accumulating bias clearly.

4.4. Robustness & Stability

The M_2 robustness margin, defined as $M_2 = 1/\|S\|_\infty$, is a measure of closed-loop sensitivity to uncertainty that is more comprehensive than gain or phase margins individually, as the M_2 margin accounts for combinations of gain and phase margin changes on system stability. The ideal value is $M_2 = 1.000$. For the H_∞ controller, $M_{2H} = 0.937066$, whereas for Youla-H $M_{2YH} = 0.937058$, a trivial 0.0008% reduction in controller robustness.

Since robust control design seeks to provide stability despite uncertainties in model dynamics, stability is likely preserved for sufficiently small perturbations in the Youla gain. However, in general, scaling the Youla parameter modifies the closed-loop pole locations and stability must be verified after gain adjustment. An easy metric is that stability is confirmed when the M_2 margin is still greater than zero, assuming the margin is already being calculated.

If a formal validation of stability post-adjustment is required, it must be shown that the new gain does not introduce right-half-plane roots in the system's characteristic polynomial for the given plant model.

4.5. Alternate System

An alternate system to demonstrate the value of the hybrid Youla-H gain tuning method is shown below in equation 12.

$$G_P = \frac{1}{s+1} \quad (12)$$

The weighting functions chosen for this system are all first-order, compared to the second-order slopes used in the drivetrain system. The weighting functions are chosen to be relaxed, allowing for easy optimization, but not so relaxed as to be unrealistic for a controller design problem. Figure 7 below shows the weighting functions as well as the resulting closed-loop transfer functions for the simple G_P under review.

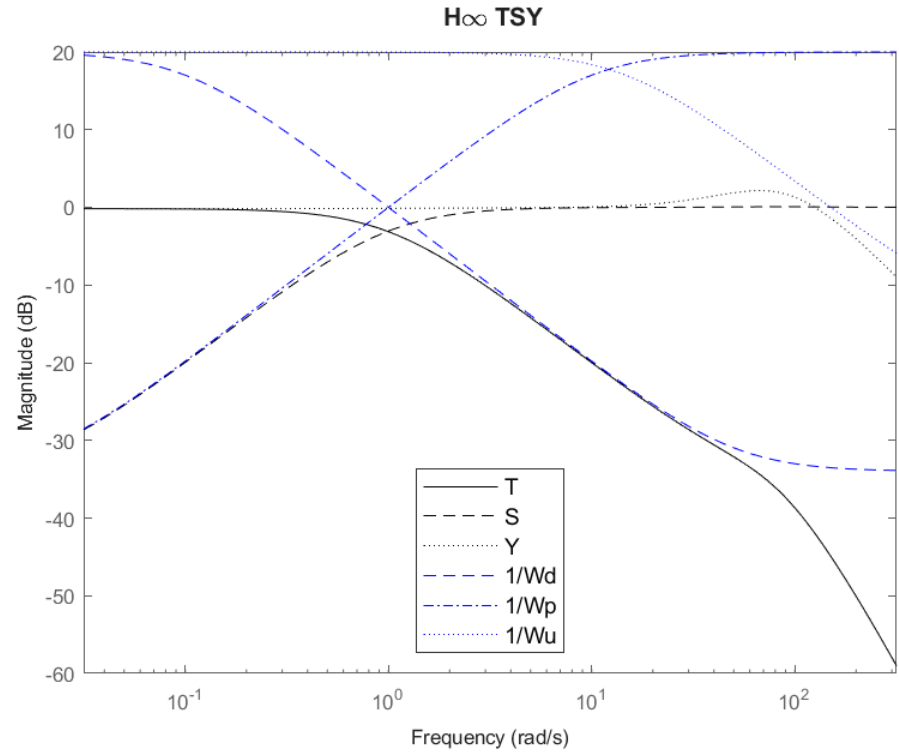


Figure 7. Weighting functions and closed-loop transfer functions for the H_∞ controller on the simple plant.

Here, $\gamma = 0.9951$. Despite the optimization succeeding, $T(0) = 0.9999817$, with a deficiency of 0.0018%. While this is approximately 1/8 of the deficiency seen for the earlier drivetrain system, it could still present an issue for long-running ramping inputs. Even with a simple plant and relaxed weights, this example serves to demonstrate that the H_∞ optimization under the best-case non-trivial example will typically still result in minor tracking deficiencies that could be integrated into noticeable tracking bias. The Youla-H method can be utilized even in this easy scenario to fix the closed-loop transfer function DC gains, particularly since code can be reused between projects with minimal modification. See Appendix A for example code.

5. Discussion

This paper presents a practical method of implementing Youla gain-tuning on H_∞ -synthesized controllers in order to strictly enforce closed-loop shape characteristics. It presents an alternative to augmenting plant dynamics prior to H_∞ synthesis and is a method that can be readily automated in controller design workflows. Due to the method's flexibility and simplicity, it can be applied by default for control designers utilizing H_∞ control methods. Whether the H_∞ solver succeeds in achieving $\gamma < 1.0$ or whether γ is reasonably close to 1.0, any time H_∞ produces a reasonable controller, the Youla-H gain tuning method can be utilized to fix the output controller. In the drivetrain system under analysis in this paper, the gain tuning resulted in a complex pole pair in the H_∞ controller

splitting into two real poles, one of which being the required pure integrator to maintain finite steady-state error under ramp inputs.

Preconditions of the Youla-H method are that the plant have a non-zero DC-gain and that the Y transfer function has a non-zero DC-gain. The primary alternative to the Youla-H method would be to augment the plant with an integrator such that the H_∞ algorithm solves for $T(0) = 1$ exactly. Due to plant augmentation requiring changes to the plant, it is a less convenient method compared to post-synthesis tuning via Youla methods.

Future work may include investigation into the limits of the Youla-H method, thorough stability analyses, and additions to the types of post-synthesis adjustments that Youla methods can make possible other than simple gain tuning. In the future, an automated controller tuning pipeline could be developed using the Youla-H method as a building block.

Author Contributions: Conceptualization, R.T., S.Y. and F.A.; methodology, R.T., S.Y. and F.A.; software, R.T. and S.Y.; validation, R.T. and S.Y.; formal analysis, R.T.; investigation, R.T.; resources, F.A.; data curation, R.T.; writing—original draft preparation, R.T.; writing—review and editing, R.T., F.A. and S.Y.; visualization, R.T.; supervision, F.A.; project administration, F.A.; funding acquisition, F.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

- MDPI Multidisciplinary Digital Publishing Institute
- DOAJ Directory of open access journals
- DC Direct Current

Appendix A Implementation

MATLAB R2024b was utilized for control design. The code below is an example demonstrating a highly automatable implementation of the Youla-H method using the plant transfer function in Equation 6. The code starts with H_∞ implementation then performs the Youla-H post-synthesis gain adjustment.

```

%% H-infinity controller implementation
% Plant and weighting function transfer functions
Gp = 22.624/(s+4.32e-5); % Plant transfer function

% makeweight(DC gain, crossover freq, high-freq gain, sample time,
% slope order)
Wd = makeweight(0.707, 18, 1e3, 0, 1);
Wp = makeweight(1e4, 18, 0.707, 0, 2);
Wu = makeweight(0.707, 300, 1e4, 0, 2);

% Perform H-inf controller synthesis
Ph = augw(Gp,Wp,Wu,Wd); % Augmented system for hinfsyn input
[sysGc, Tzw, gamma] = hinfsyn(Ph); % Perform H-infinity synthesis

mreal_tol = 0.001; % Tolerance for minreal for pole/zero pairs
[Gc_n,Gc_d] = tfdata(sysGc); % Reformat output of hinfsyn
Gc_H = zpke(minreal(tf(Gc_n,Gc_d),mreal_tol)); % Hinf Gc

% Construct closed-loop transfer functions
L_H = Gp*Gc_H;
S_H = zpke(minreal(1/(1+L_H),mreal_tol));
T_H = zpke(minreal(L_H/(1+L_H),mreal_tol));
Y_H = zpke(minreal(Gc_H/(1+L_H),mreal_tol));
T0_H = evalfr(T_H,0); % Evaluate T(0)

%% Perform Youla-H post-synthesis gain adjustment
syms K, s
[n,d] = tfdata(Gp);
n = cell2mat(n); d = cell2mat(d); % Reformat output
Gp_sym = poly2sym(n,s)/poly2sym(d,s); % Construct symbolic Gp

[n,d] = tfdata(Y_H);
n = cell2mat(n); d = cell2mat(d); % Reformat output
Y_YH = K*poly2sym(n,s)/poly2sym(d,s); % Construct symbolic Y_YH
T_YH = simplifyFraction(Y_YH*Gp_sym); % Construct symbolic T_YH

% Perform gain adjustment
K = solve(subs(T_YH,s,0)-1, K); % Solve for new gain K
Y_YH = subs(Y_YH); % Insert solved K

% Convert back to transfer functions
[n,d] = numden(Y_YH);
Y_YH = zpke(tf(sym2poly(n),sym2poly(d)));
Gc_YH = zpke(minreal(Y_YH/(1-Y_YH*Gp),mreal_tol));
T_YH = zpke(minreal(Gc_YH*Gp/(1+Gc_YH*Gp),mreal_tol));
S_YH = zpke(minreal(1/(1+Gc_YH*Gp),mreal_tol));
T0_YH = evalfr(T_YH,0); % Evaluate new T(0)

```

References

1. Zhou, K.; Doyle, J.C.; Glover, K. *Robust and Optimal Control*; Prentice Hall: Upper Saddle River, NJ, USA, 1996.
2. Skogestad, S.; Postlethwaite, I. *Multivariable Feedback Control: Analysis and Design*, 2nd ed.; Wiley: Chichester, UK, 2005.
3. Assadian, F.; Mallon, K.R. *Robust Control: Youla Parameterization Approach*; John Wiley & Sons: Hoboken, NJ, USA, 2022.
4. Tan, R.; Yadav, S.; Assadian, F. Systemic scaling of powertrain models with Youla and $\mathcal{H}_\infty$ driver control. *Energies* **2025**, *18*, 3126. <https://doi.org/10.3390/en18123126>
5. Seatzu, C.; Usai, E.; Zaccarian, L. H-infinity controllers with integral action: an experimental evaluation. *IFAC Proceedings Volumes*, 2008, 41(2), pp. 14,916–14,921.
6. Xiaohai Hu, Jason Laks, Guoxiao Guo, Xu Chen, Iterative Youla-Kucera Loop Shaping For Precision Motion Control. *IFAC-PapersOnLine*, Volume 59, Issue 30, 2025, pp. 575-580, ISSN 2405-8963

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